

solutions for devices based on the IoT, augmented and virtual reality, and voice-based devices like Amazon Echo Dot and Google Home create new experiences and solutions for clients' unique business needs."

Since data is stored on external servers, companies need to be sure that it is safe and cannot be accessed by unauthorised parties. There should be a high-speed Internet connection as slow connections can hamper performance.

Some vendors have developed offline functionality to allow people to keep working even if the Internet goes down. All data is synced to the system once a good connection is available again. In SaaS, businesses rely on outside vendors for software and its updates, and facilitate a secure environment for their business data. They should understand their SaaS provider's SLA and make sure it is enforced.

Software-defined networking

Software-defined networking (SDN) defines how a networking and computing system can be built using a combination of open, software-based technologies and commodity networking hardware. It enables engineers and administrators of the cloud computing network to respond quickly to changing business requirements via a centralised control console.

SDN encompasses multiple kinds of network technology designs. This makes the network more flexible and agile to support the virtualised server and storage infrastructure of the modern data centre. SDN is an approach to design, build and manage a network that separates its control or SDN policy (brains) and forwarding (muscle) planes. This enables network control to become directly programmable and underlying infrastructure

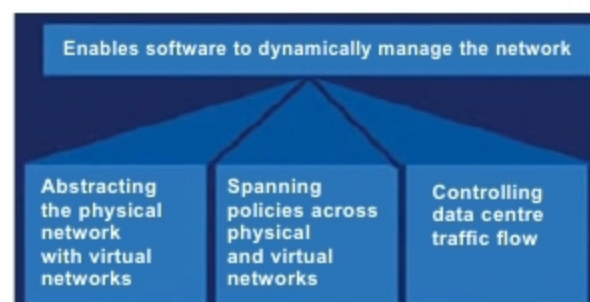


Fig. 2: Software-defined networking (Credit: <https://blog.machsol.com>)

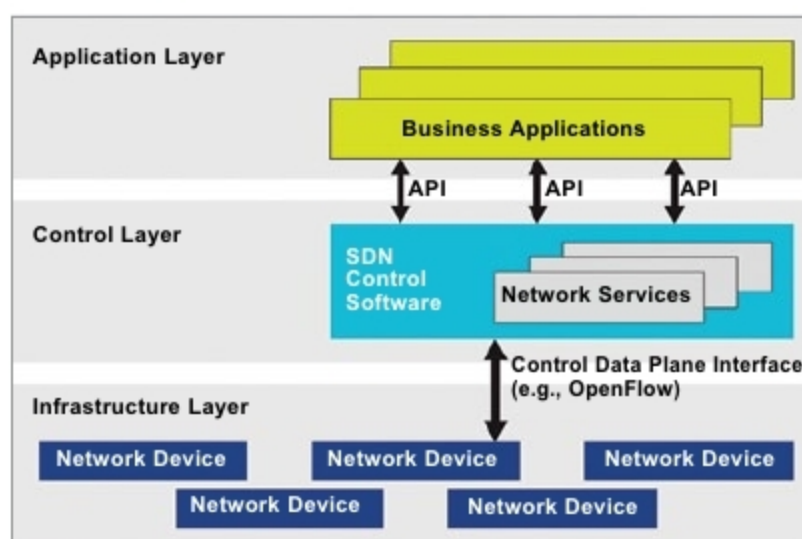


Fig. 3: SDN architecture (Credit: www.sdxcentral.com)

to be abstracted for applications and network services for applications such as SDN cloud computing or mobile networks.

The goal of SDN is to improve network control to enable enterprises and service providers to respond quickly. It is an emerging architecture that is dynamic, cost-effective, manageable and adaptable. It is ideal for the high-bandwidth and dynamic nature of today's applications.

The foundational element for building SDN solutions is OpenFlow protocol. A network engineer can shape traffic from a centralised control console in SDN without working on individual switches in the network.

Software-defined wide-area network (SD-WAN) is an extension of SDN. This technology allows an enterprise to connect all their networks within their system across a wide geographic area. For example, all data centres and branch office networks within an enterprise are connected via SD-WAN. Its ability to unify various connections are turning organisations and companies towards SD-WAN.

Another benefit of SD-WAN is end-to-end encryption across the network, which increases security.

SDN architecture to support SaaS

SDN removes the need to purchase application-specific integrated circuit (ASIC)-based networking hardware, and supports pay-as-you-go models. This reduces operational expenses.

SDN enables algorithmic control of the network of programmable elements. This makes it easier to design, deploy, manage and scale networks.

The ability to automate optimises service availability and reliability by reducing overall management time and chance for human error.

SDN helps organisations deploy new applications, services and infrastructure to quickly meet changing business goals and objectives. It enables them to implement new business models that can offer

new revenue streams and more value from the network.

SDN has the potential to revolutionise legacy data centres by providing a flexible way to control the network. It provides different services to different organisations. For cloud computing and data centres, network virtualisation offers better utilisation of resources and faster turnaround times for creating a segregated network.

Conclusion

SaaS refers to business software applications delivered via the cloud. Given the widespread cloud accessibility, it is easier, faster and less expensive to roll out applications. Today, nearly every type of core business function is available via SaaS.

SDN enables the programming of network behaviour in a centrally-controlled manner using open APIs. By implementing SDN control layer, operators can manage the entire network and its devices consistently, regardless of the complexity of the underlying network technology.

The shift to a software-centric model is disrupting traditional industrial markets and companies. Many manufacturers are starting on a path to fundamentally transform their business models and cultures by forging digital connections with suppliers, customers and other manufacturers, and by creating digital-based products and services that deliver business outcomes. As industries embrace new technologies and innovate to improve operations and solve customer problems, the move to a software-centric model will drive digital transformation. **EFY**